

Less is More: Updating 3% of Attention Heads Reduces Hallucination in LLaVA by 38%

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Overview

Our goal is to reduce object hallucination in LLaVA-1.5-7B without expensive full-model retraining.

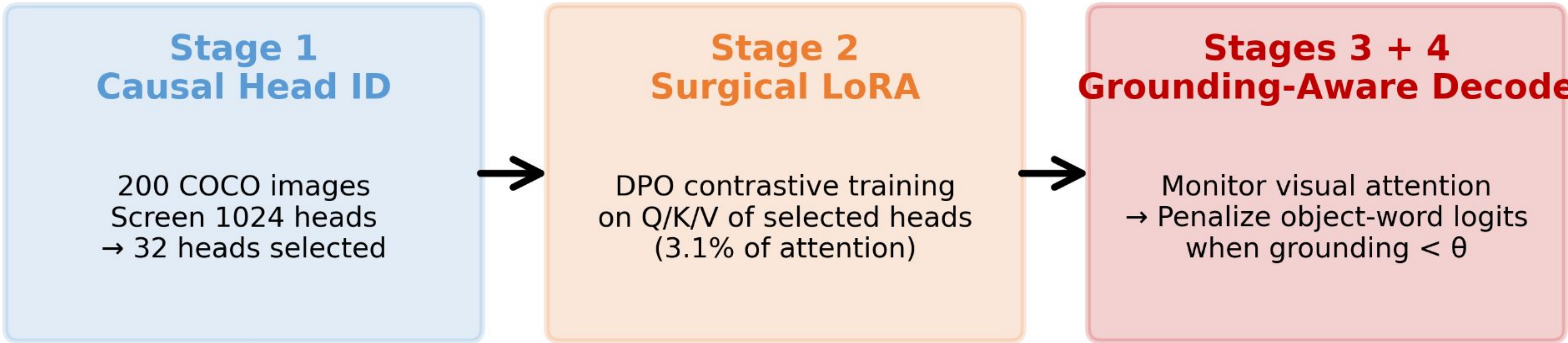
Our approach: combine surgical LoRA on causally-identified attention heads + inference-time grounding controller that penalizes hallucinated content words when visual attention is weak.

Dataset & Evaluation

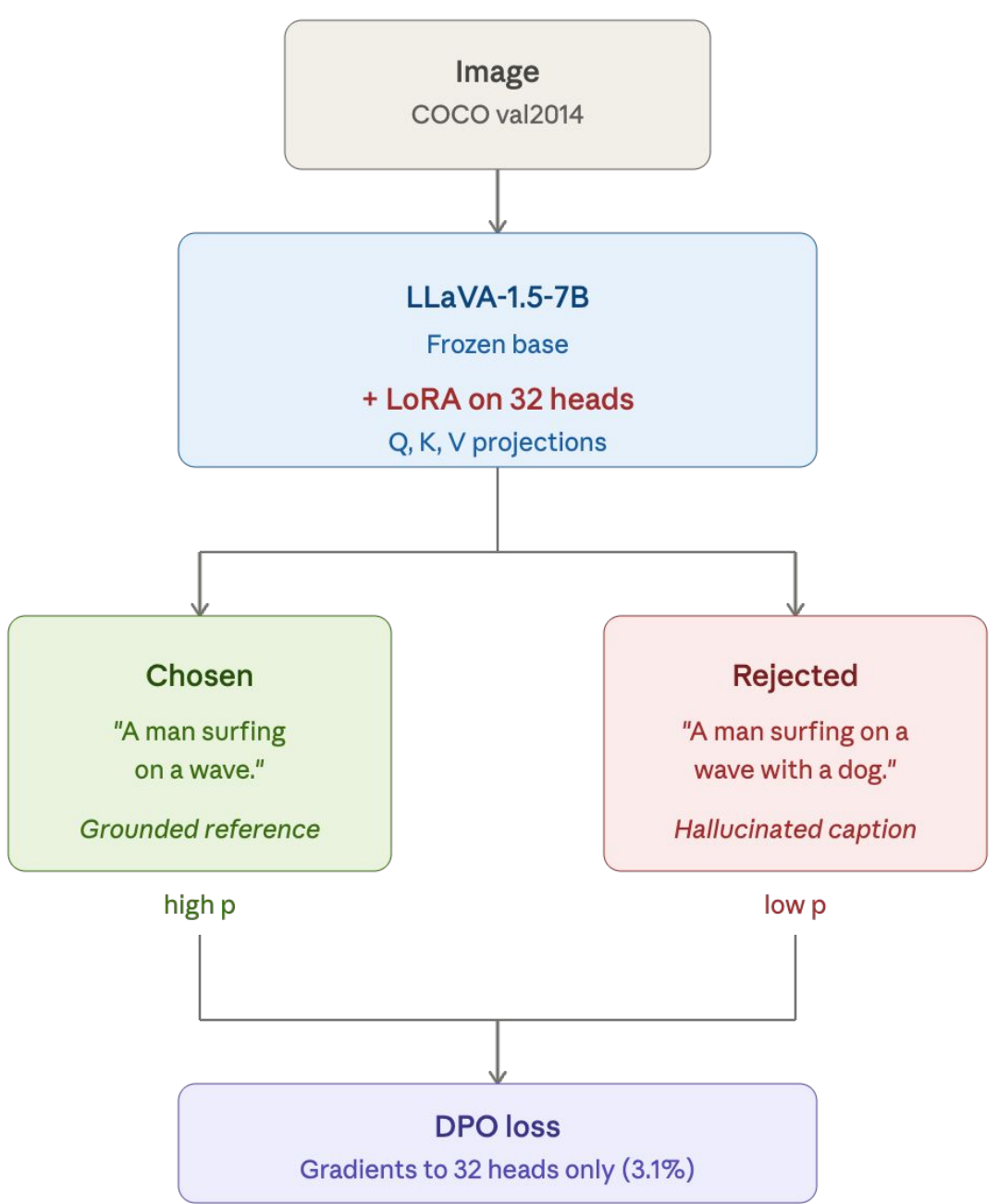
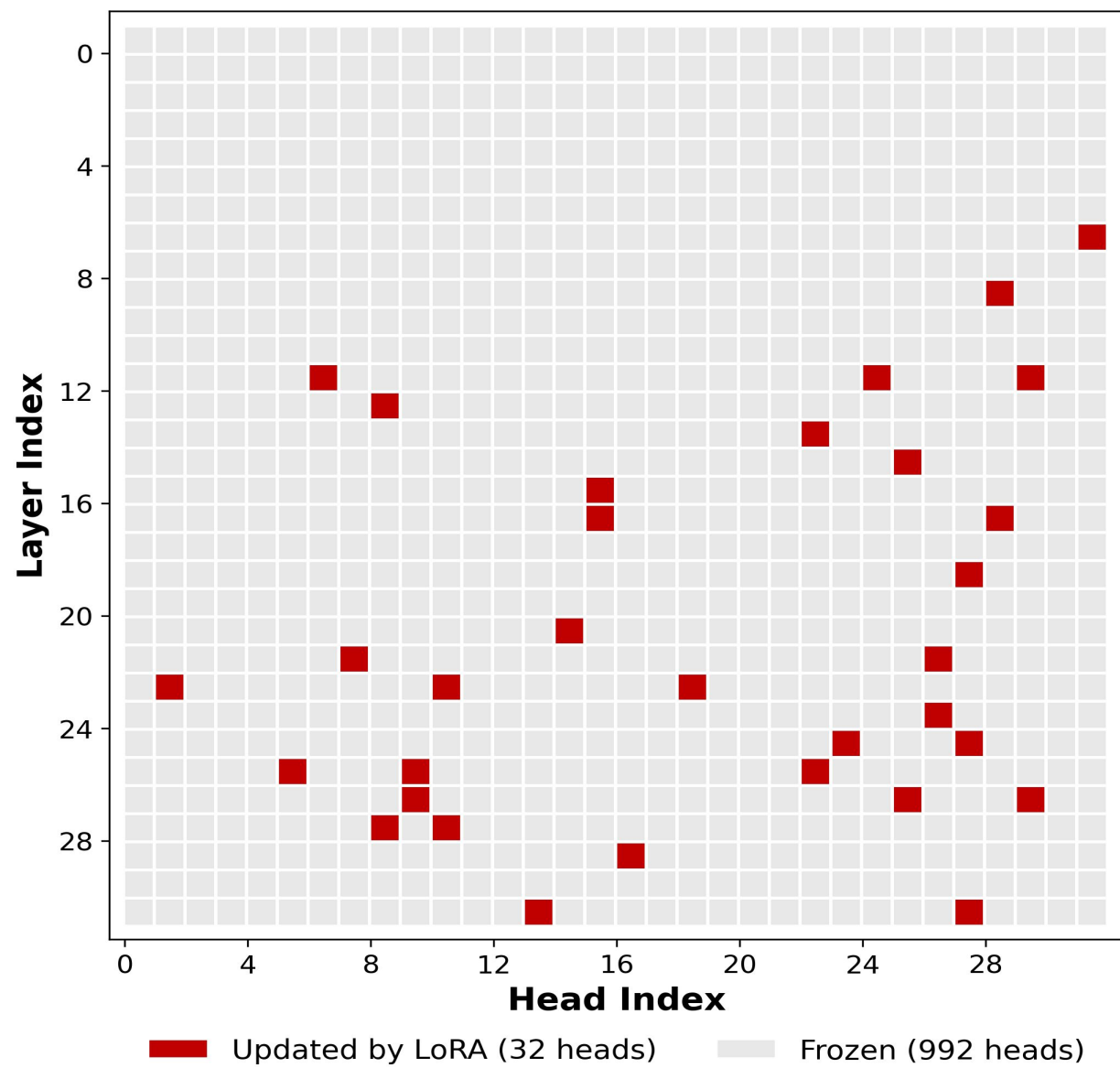
- COCO val2014. 200 images for Stage 1 (head identification). 400 held-out images for evaluation.
- Primary metric: CHAIR (Rohrbach et al., 2018).
- Side metrics: caption length, GT-object recall.

Method

- LLaVA-1.5-7B as base model. Greedy decoding baseline.
- Hypothesis: targeted intervention on a small set of causally-identified attention heads is more effective than blanket fine-tuning.

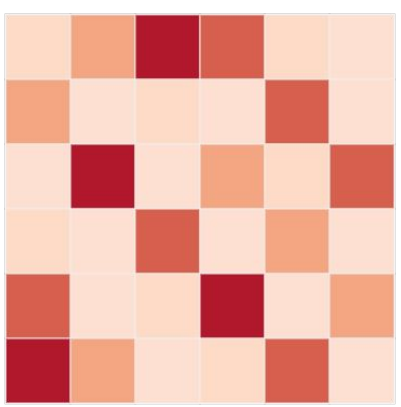


Surgical LoRA — 32 of 1024 Attention Heads (3.1%)



$$L = -\log \sigma(\beta \cdot [(\log p_{\theta}(\text{chosen}) - \log p_{\text{ref}}(\text{chosen})) - (\log p_{\theta}(\text{rejected}) - \log p_{\text{ref}}(\text{rejected}))])$$

Visual attention at decode step t



Grounding score

$$g_{\text{t}} = \text{mean}_H \left(\frac{\sum \alpha_{h,i}}{\sum \alpha_{h,j}} \right)$$

$i \in \text{img}$
Mean visual attention mass over 32 heads

if $g_{\text{t}} < \theta$

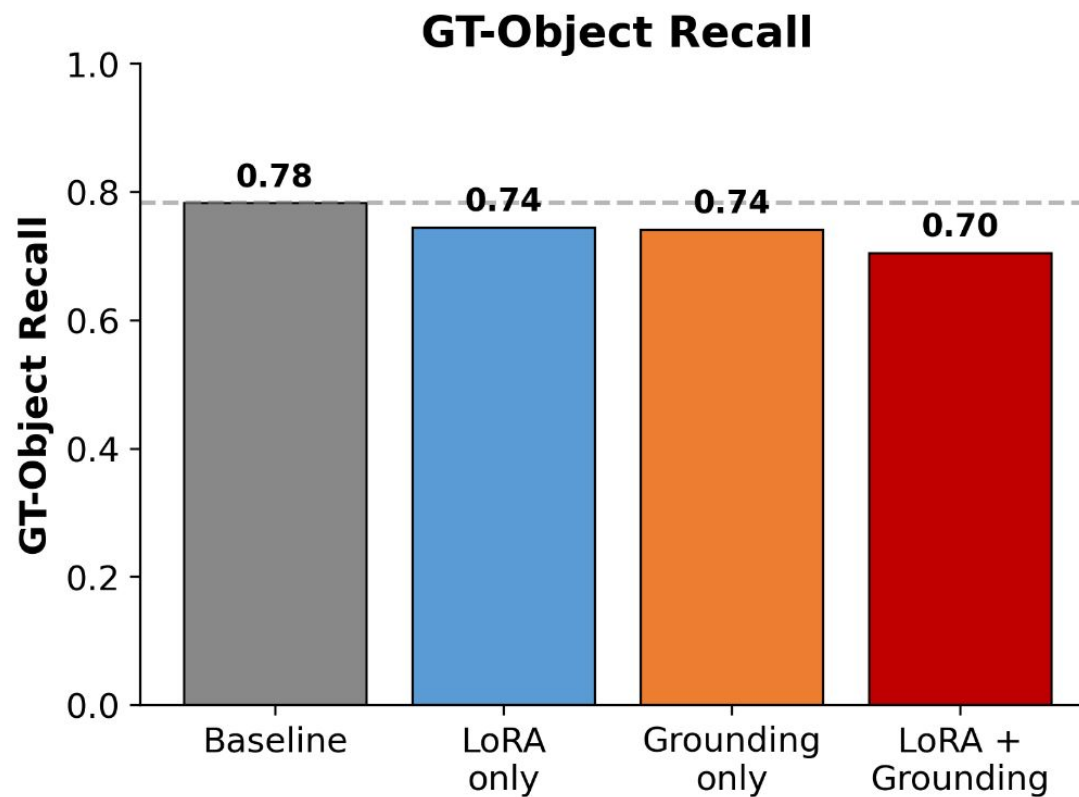
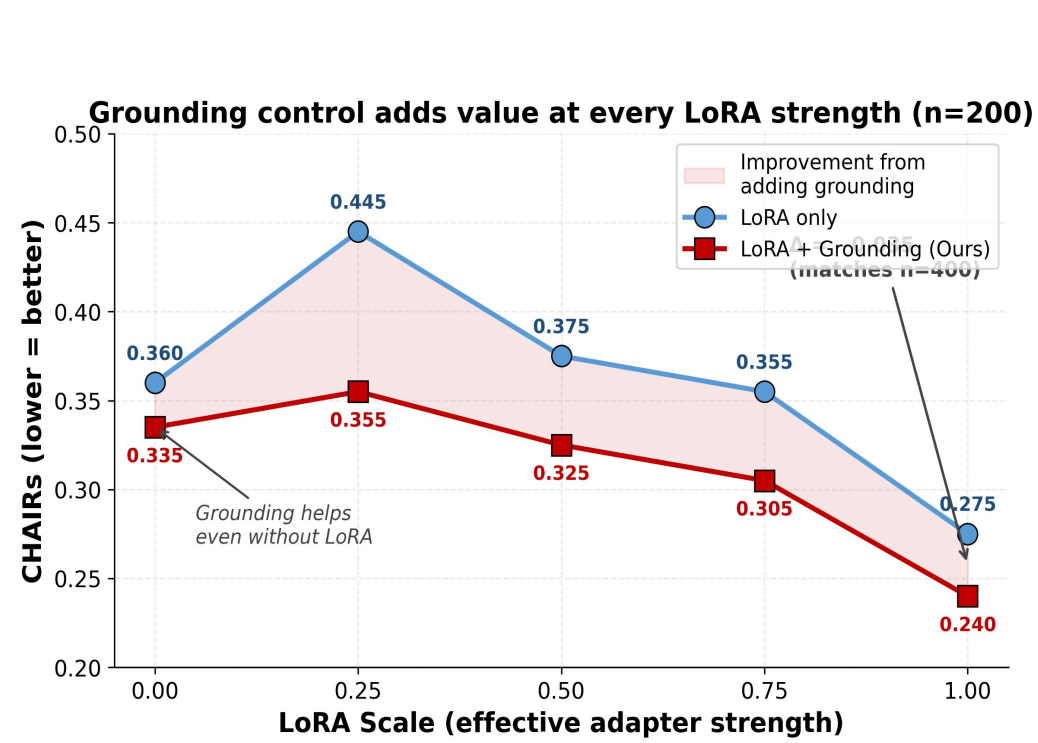
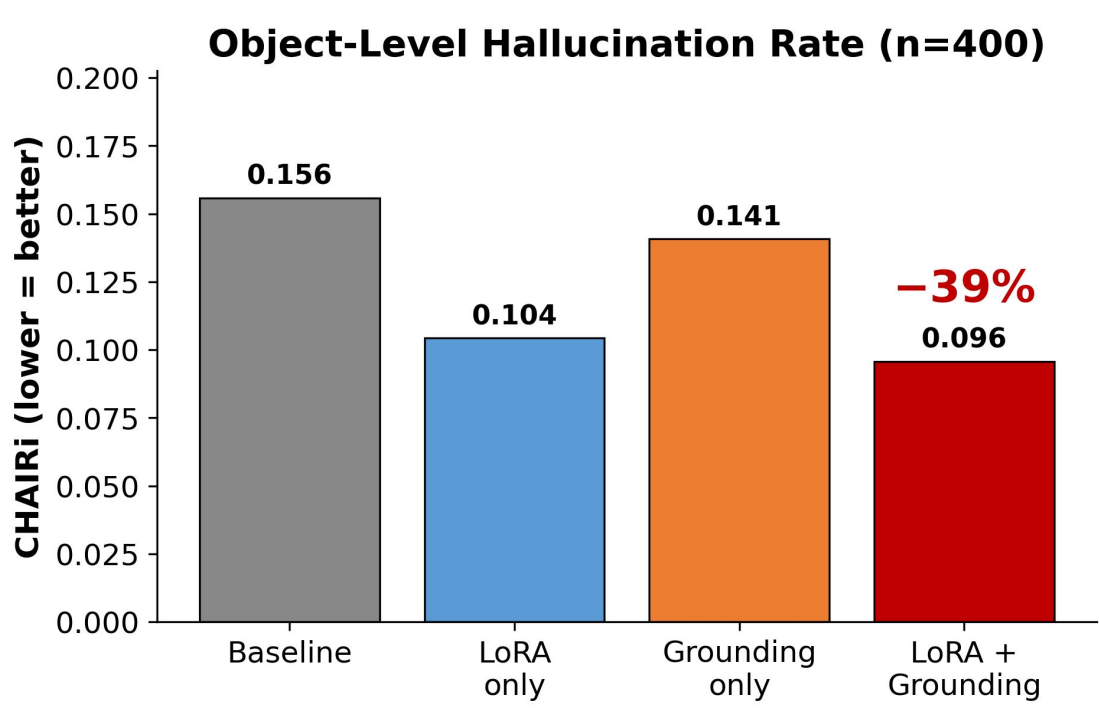
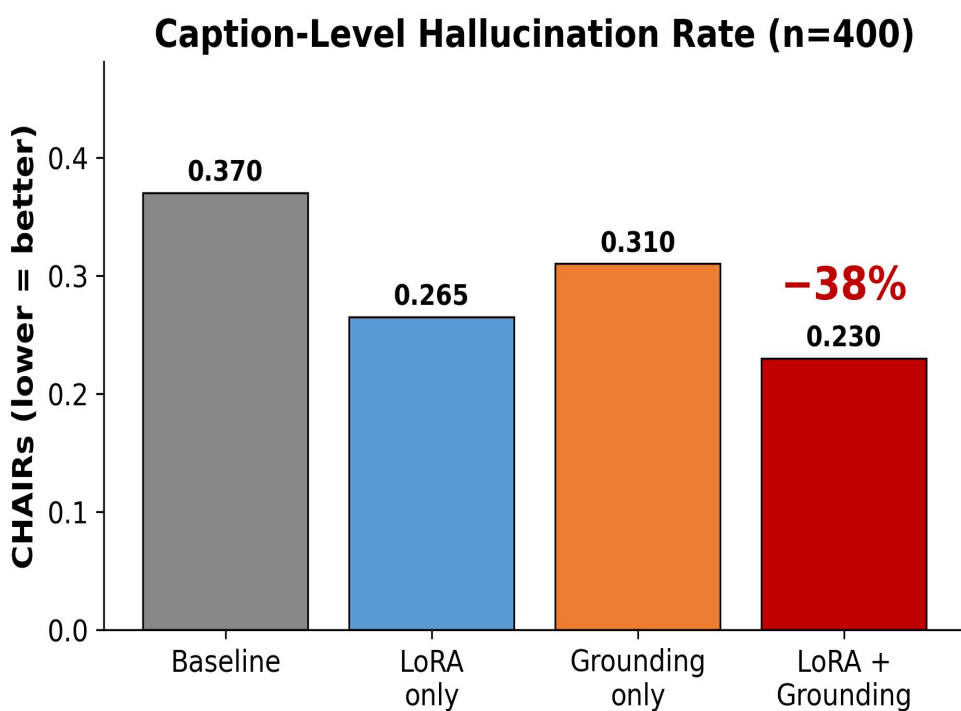
Logit penalty

$$\ell_{\text{v}} \rightarrow \ell_{\text{v}} + \alpha \cdot (g_{\text{t}} - \theta)$$

For each object-word token v
(nouns + adjectives in COCO vocab)

Example penalized tokens:
"dog", "table", "person",
"car", "umbrella", ...

Results



Baseline
The image showcases a large, clean kitchen with wooden cabinets and granite countertops. A **refrigerator** is situated on the right side of the room, while an **oven** can be found near the center-left area.

LoRA only
A kitchen with a **refrigerator** and **sink**. A brown counter top, cabinets and a window above the sink. A green **bottle** on the countertop.

Grounding only
The image showcases a large, clean kitchen with wooden cabinets and granite countertops. A **refrigerator** is situated on the right side of the room while an **oven** can be found in front of it. There are t

LoRA + Grounding (Ours)
A kitchen with a **refrigerator** and **sink**. A brown counter top in front of the **fridge**, cabinets above it, a window behind the stove topside down on its side.

Conclusion

Surgical LoRA + grounding control gives **-38% CHAIR** on 400 COCO images, updating only 3.1% of attention.

Trade-off: Captions become more concise (78% → 70% recall), a favorable precision/recall shift for safety-critical use.